SEMINAR REPORT
ON

Loan Default Risk Prediction using Random Forest

BY

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CERTIFICATE

This is to certify that the seminar report entitled **"Loan Default Risk Prediction using Random Forest"** submitted by Prasad Bhalerao (PRN No. RBT24CB026) is a record of bonafide work carried out under my supervision, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Business Systems of Savitribai Phule Pune University, during the academic year 2026-27.

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Abstract

This seminar studies the Random Forest algorithm applied to credit risk, and reports a complete working implementation. The dataset is the public Kaggle “Give Me Some Credit” collection of 150,000 labelled borrower records with 10 numeric attributes. The target, SeriousDlqin2yrs, marks borrowers 90 or more days past due within two years — only 6.68% of records.

That imbalance makes accuracy unusable as a headline metric: always predicting the majority class scores 93.3% while catching nobody. The work therefore selects on the area under the precision-recall curve, treats undocumented placeholder codes explicitly, and chooses the decision threshold on validation data rather than assuming 0.50.

On 30,000 held-out records scored exactly once, the model reached ROC-AUC 0.8654 and PR-AUC 0.4014 against a 0.0668 no-skill baseline. The pipeline was deployed as a REST service with a web interface. The report also documents the limitations that make the system unfit for real credit decisions.

Keywords: Random Forest, Credit Risk Prediction, Class Imbalance, Precision-Recall Curve, Decision Threshold, Supervised Classification

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CHAPTER 1: INTRODUCTION

1.1 Overview

Credit risk prediction estimates how likely a borrower is to miss repayment. Traditionally an underwriter applied policy rules by hand. Machine learning instead fits a model to past borrowers whose outcomes are known, and learns the link between attributes and outcome.

This seminar studies that application through the Random Forest algorithm. Each record describes one borrower by 10 numeric attributes; the label states whether they were 90+ days past due within two years. It is supervised binary classification — the output is a category, not a quantity, and the groups come from labels rather than being discovered.

1.2 Need and Importance

Lenders extend credit under uncertainty. A default is a direct loss; a sound applicant wrongly refused is lost revenue. Better ranking of applicants by risk therefore has direct economic value, and manual assessment does not scale.

The problem is also statistically interesting. Only 6.68% of borrowers here are positive, and that imbalance breaks the most familiar metric: a model that always answers “no delinquency” scores 93.3% accuracy while catching nobody. Which metrics replace accuracy, and why, is a central theme of this report.

1.3 Objectives

- To understand the Random Forest algorithm: bagging, random feature selection, and how an ensemble yields a probability.
- To study the architecture of a complete prediction system, from raw data to a deployed service.
- To examine correct evaluation under class imbalance, and show why accuracy is inadequate.
- To implement the method on a real dataset and report measured results.
- To identify the advantages, limitations and ethical challenges of such systems.

CHAPTER 2: LITERATURE REVIEW

Breiman [1] introduced Random Forests: combining decorrelated trees reduces variance without raising bias. Two sources of randomness are responsible — bootstrap sampling of rows, and restricting each split to a random feature subset.

Saito and Rehmsmeier [4] were the most influential source here. ROC plots mislead under imbalance, because the false-positive rate is measured against a large negative population and barely moves; precision-recall plots do not, since precision counts only what was flagged. This is why the model below is selected on PR-AUC. Pedregosa et al. [2] document scikit-learn, whose forest averages per-tree leaf proportions rather than taking the majority vote introductory treatments describe. Chawla et al. [5] frame imbalance as a choice of operating point on a trade-off curve.

Table 2.1: Summary of literature review

Sr.	Author(s), Year	Focus	Key Findings
1	Breiman, 2001	Random Forest theory	Averaging decorrelated trees cuts variance; error converges as trees are added.
2	Pedregosa et al., 2011	scikit-learn	Reference implementation. Probability = mean of per-tree leaf proportions, not a vote.
3	Kaggle, 2011	Give Me Some Credit data	150,000 labelled records with a documented data dictionary.
4	Saito & Rehmsmeier, 2015	Imbalanced evaluation	ROC is over-optimistic under imbalance; PR is more informative.
5	Chawla et al., 2002	Resampling, cost-sensitive learning	Imbalance is a choice of operating point; reweighting shifts it rather than improving ranking.

Research gap.

The literature agrees that accuracy and ROC-AUC flatter a classifier when positives are rare, yet both are still widely reported, and the standard remedies are often applied without checking whether they improve ranking or merely relocate the operating point. This work measures that: class weighting moves the model along the precision-recall curve without lifting it, so explicit threshold selection is used instead. Separately, undocumented placeholder codes that much published work imputes away are shown to carry real signal.

CHAPTER 3: TECHNOLOGY / CONCEPT OVERVIEW

3.1 Basic Concepts

Table 3.1: Key concepts and definitions

Term	Definition
Decision tree	Splits data repeatedly on one attribute. Leaves hold the training samples that reached them.
Gini impurity	$G = 1 - \Sigma\, pk^2$, where pk is the proportion of class k in a node. Splits maximise its weighted reduction.
Bagging	Each tree trains on a sample drawn with replacement, so averaging their output reduces variance.
Random Forest	An ensemble of 300 such trees, each split restricted to a random feature subset.
Precision / Recall	Of those flagged, the share truly delinquent — TP/(TP+FP). Of the truly delinquent, the share found — TP/(TP+FN).
F1-score	Harmonic mean 2PR/(P+R); dominated by the weaker term.
ROC-AUC	Probability a random positive outranks a random negative. 0.5 is chance.
PR-AUC	Area under the precision-recall curve. No-skill baseline = the positive rate, here 0.0668.

3.2 Architecture and Working

Training, evaluation and figure generation run offline and produce one saved artifact. The web service loads it at startup and scores requests; it never trains. The browser talks to one origin only — requests to /api are forwarded server-side, so no cross-origin request is ever made.

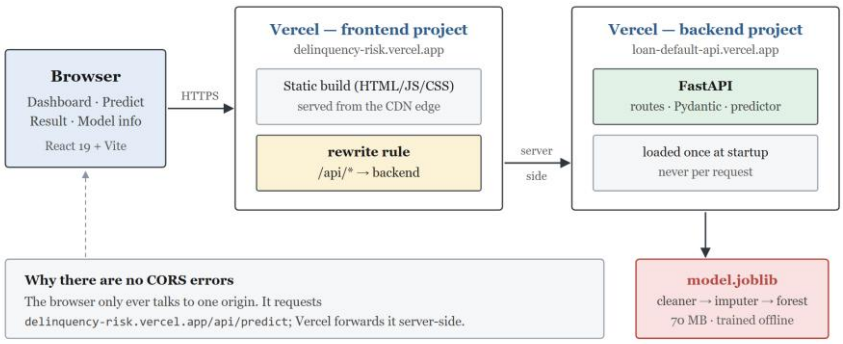


Figure 3.1: System architecture of the implemented application

Training runs in six stages: load and validate against a schema; mark implausible values missing; split 60/20/20 stratified; compare five candidates on validation; select the threshold; score once on test. Every fitted quantity — imputation medians, model parameters, the threshold — comes from train and validation only. Stratifying matters

at a 6.68% base rate, or the three parts would carry different class ratios and their metrics would not be comparable.

To predict, each of the 300 trees routes the record to a leaf and reports that leaf's positive proportion. The forest averages those 300 values — scikit-learn averages per-tree probabilities rather than taking a majority vote — and the mean is compared with the threshold of 0.2302 to give a class.

3.4 Features and Data Quality

Ten numeric attributes per borrower. No categorical encoding is needed, and because trees split on rank order, no scaling is either.

Table 3.2: Input features

Feature	Description
RevolvingUtilizationOfUnsecuredLines	Balances ÷ total credit limits
age	Borrower age in years
NumberOfTime30-59DaysPastDue	30–59 day late episodes, last 2 years
DebtRatio	Debt, alimony and living costs ÷ gross monthly income
MonthlyIncome	Gross monthly income (19.82% missing)
NumberOfOpenCreditLinesAndLoans	Open loans and credit lines
NumberOfTimes90DaysLate	90+ day late episodes
NumberRealEstateLoansOrLines	Mortgages and home equity lines
NumberOfTime60-89DaysPastDue	60–89 day late episodes, last 2 years
NumberOfDependents	Dependents excluding the borrower (2.62% missing)

Table 3.3: Data quality issues and treatment

Issue	Extent	Treatment
MonthlyIncome missing	29,731 (19.82%)	Median imputation + missing indicator
NumberOfDependents missing	3,924 (2.62%)	Median imputation + missing indicator
Counters coded 96 or 98	269 rows	Placeholders. Imputed with an indicator — these rows are 54.65% positive against a 6.68% base rate
Age recorded as 0	1 row	Impossible; treated as missing
Duplicate predictor rows	646 rows	Retained — plausibly distinct borrowers
Extreme ratios	up to 329,664	Retained — trees split on rank order

Preprocessing and the classifier are one scikit-learn Pipeline, and that composite object is what is saved. Serving code loads and calls it, so train-serve skew is structurally impossible. The classifier receives 16 columns built from the 10 features, because of the appended indicators.

3.5 Model Selection

Five configurations were fitted on train and compared on validation, ranked by PR-AUC.

Table 3.4: Validation comparison of configurations

Configuration	ROC-AUC	PR-AUC	Precision	Recall
baseline	0.8450	0.3481	0.5334	0.1870
regularised	0.8615	0.3820	0.5988	0.1556
balanced	0.8602	0.3776	0.2399	0.7197
balanced subsample	0.8602	0.3780	0.2805	0.6504
regularised with indicators	0.8630	0.3863	0.6290	0.1581

Raising the minimum leaf size to 20 lifted PR-AUC from 0.3481 to 0.3820 — the unrestricted forest was overfitting.

Class weighting raised recall from 0.1556 to 0.7197 but left PR-AUC flat at 0.3776. It moved the model along the curve without lifting it, so it was not used — a threshold does the same job visibly and without refitting.

Adding missing-value indicators won, at 0.3863. The threshold was then set on validation by maximising F1, giving 0.2302 rather than 0.50. The model was refitted on train + validation and scored once on test.

CHAPTER 4: APPLICATIONS AND RESULTS

4.1 Applications and Use Cases

Application scoring gives a risk score at origination, combined with policy rules and affordability checks, so every applicant is assessed on the same basis in milliseconds. **Behavioural scoring** rescores an existing portfolio as payment behaviour accumulates, catching accounts drifting toward delinquency early enough to intervene. **Collections prioritisation** ranks delinquent accounts so finite recovery effort goes where it matters; it needs only the ranking, not a calibrated probability.

The technique generalises to any rare-event problem on tabular data — fraud detection, claim triage, churn, predictive maintenance. The benefits are consistency, scale and inspectability: identical criteria for every applicant, at volumes no manual process could reach.

4.2 Deployed Application

The model is deployed at https://delinquency-risk.vercel.app, with API documentation at https://loan-default-api.vercel.app/docs. Four pages — Dashboard, Predict, Result and Model info — all fetching their values from the service at run time. The dashboard states what the model does and does not do, and shows the majority-class baseline beside the accuracy so the latter cannot be mistaken for a strong result.

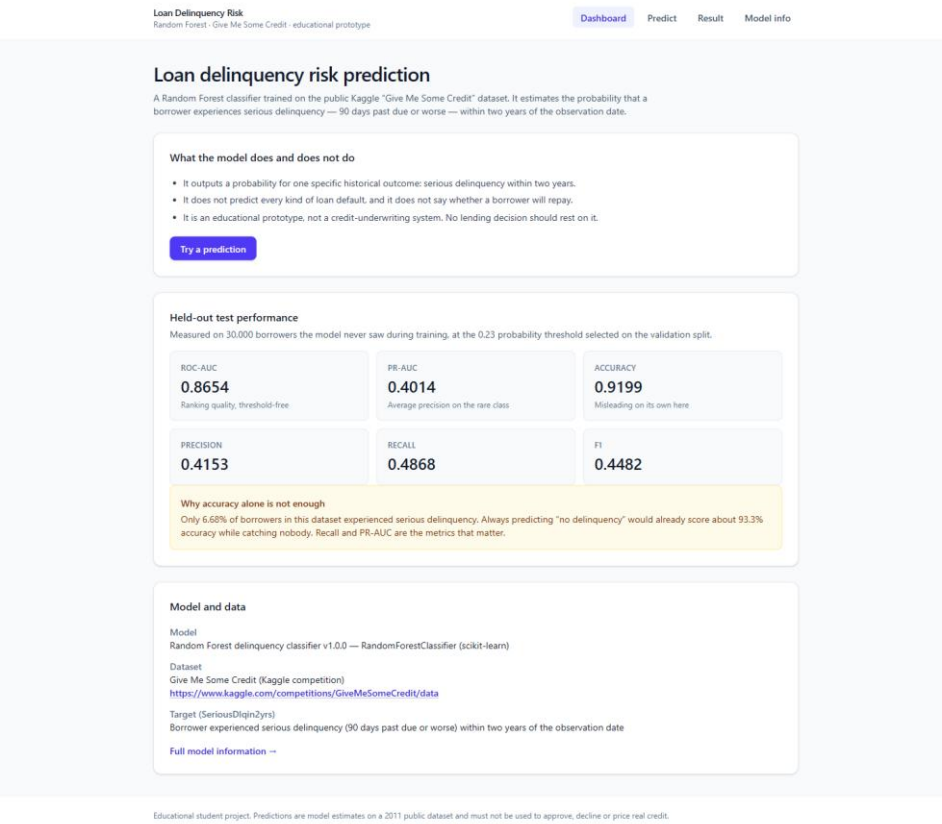


Figure 4.1: Dashboard of the deployed application

4.3 Results

Measured on 30,000 held-out borrowers, scored once, at threshold 0.2302.

Table 4.1: Held-out test metrics

Metric	Value	Read it as
ROC-AUC	0.8654	Ranks a delinquent above a sound borrower 86.5% of the time
PR-AUC	0.4014	~6× the no-skill baseline of 0.0668
Recall	0.4868	Caught 976 of 2,005 real delinquencies
Precision	0.4153	42% of those flagged truly were
F1-score	0.4482	Positive class
Accuracy	0.9199	Below the 93.3% of always answering “no”

Table 4.2: Per-class report and confusion matrix

Class	Precision	Recall	F1	Support
0 — No delinquency	0.9628	0.9509	0.9568	27,995
1 — Delinquency	0.4153	0.4868	0.4482	2,005
Macro average	**0.6891**	**0.7189**	**0.7025**	**30,000**
Weighted average	0.9262	0.9199	0.9228	30,000
Confusion matrix	TN 26,621	FP 1,374	FN 1,029	TP 976

This is the imbalance problem in one table: 0.9568 F1 on the majority class against 0.4482 on the minority. The weighted average of 0.9228 looks impressive but is

dominated by 27,995 negatives. Quote the macro average (0.7025) or the positive-class F1 — never the weighted one.

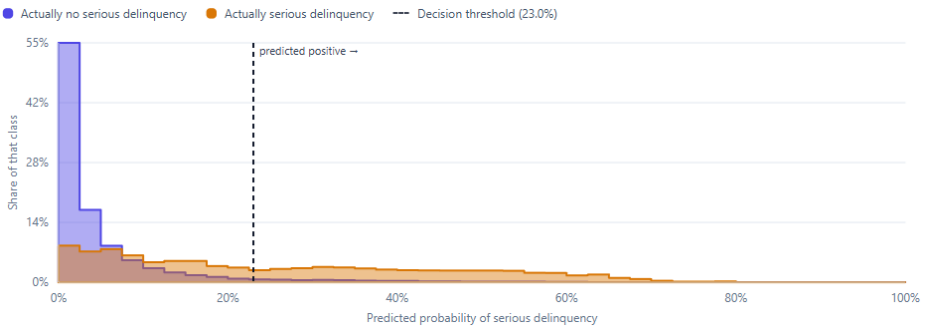


Figure 4.2: Predicted score distribution by actual outcome

Most borrowers who stayed current score near zero, but those who became delinquent are spread across the whole range rather than clustered high. That overlap — not a defect in the implementation — is what limits recall.

Table 4.3: Effect of the decision threshold

Threshold	Accuracy	Precision	Recall	F1	Flagged
0.1000	0.8469	0.2610	0.7047	0.3810	5,413
0.2000	0.9131	0.3884	0.5222	0.4454	2,696
0.2302	0.9199	0.4153	0.4868	0.4482	2,350
0.3000	0.9292	0.4650	0.3970	0.4283	1,712
0.5000	0.9374	0.6267	0.1566	0.2506	501

The same model behaves very differently by cut-off: at 0.10 recall exceeds 0.70 but three in four flags are false alarms; at 0.50 precision passes 0.62 but only one delinquency in six is found. Accuracy rises monotonically simply because the model answers “no” more often — a further reason not to trust it here.

Five-fold cross-validation on train + validation, test excluded throughout, gives ROC-AUC 0.8632 ± 0.0053 and PR-AUC 0.3978 ± 0.0148; the test values sit within one standard deviation of both, so the result is not a lucky split. ROC-AUC does fall from 0.9329 in-sample to 0.8654 held-out, a gap of 0.0675 — some overfitting remains despite regularisation, and is reported rather than concealed.

CHAPTER 5: ADVANTAGES AND LIMITATIONS

5.1 Key Limitations

Data, not algorithm. US consumers around 2011, with no date column — so no out-of-time validation is possible and the sampling frame is undocumented.

Uncalibrated. A predicted 0.30 is a ranking score, not a claim that 30 in 100 such borrowers default.

Fairness untested. No race, sex or geography in the data — but that is not protection, since age is used and income can proxy for protected traits. Absence of measured bias is absence of measurement.

Threshold economics. Maximising F1 weights a missed default and a false alarm equally; real lending does not.

5.2 Challenges Encountered

Undocumented codes. The codes 96 and 98 appear in 269 records with no explanation. Those rows are 54.65% positive against a 6.68% base rate, so imputing them away would discard signal while asserting clean payment histories. Imputing with a retained indicator lets the model learn the value was unavailable without guessing what the codes mean.

Deployment and skew. The Python stack plus a 70 MB model is ~315 MB, so runtime and training dependencies were split to fit the 500 MB limit. Any train/inference preprocessing difference would corrupt predictions silently — one saved pipeline removes the possibility.

Table 5.1: Advantages and limitations

Advantages	Limitations
Captures non-linear interactions without feature engineering	Data describes 2011 US consumers; current performance unknown
Robust to extreme values; no scaling or encoding needed	No date column, so out-of-time validation is impossible
Outputs probabilities, not just labels	Probabilities are uncalibrated; calibration was not evaluated
Threshold adjustable after training, without refitting	Threshold optimises F1, weighting both error types equally
Reports feature importances, keeping the model inspectable	Importance describes splits, not causation
Trains in ~1 minute; reproducible with a fixed seed	No protected attributes, so fairness cannot be tested

CHAPTER 6: FUTURE SCOPE

Each item below follows directly from a limitation in Chapter 5.

- **Probability calibration.** Fit a calibrator on held-out data and report a reliability diagram and Brier score. The single change that would most improve practical usefulness.
- **Cost-sensitive thresholds.** Replace the F1 criterion with an explicit cost matrix, placing the operating point where expected loss is minimised.
- **Out-of-time validation.** With origination dates, train on an earlier period and test on a later one — standard for credit models, precisely because random splits flatter them.
- **Better explanations.** Permutation importance on held-out data is less biased than the impurity measure used here; per-prediction attributions such as SHAP would support the adverse-action explanations regulated lending requires.
- **Fairness assessment.** With the relevant attributes, test for disparate impact across protected groups and mitigate where needed.

The field is moving toward models that are auditable, not merely accurate — regulation increasingly requires that an automated credit decision be explainable to the person it affects.

CHAPTER 7: CONCLUSION

A Random Forest was trained on 150,000 labelled records and evaluated on 30,000 held-out borrowers scored exactly once, reaching ROC-AUC 0.8654 and PR-AUC 0.4014 against a 0.0668 baseline.

The methodological findings matter as much as the numbers:

- Accuracy is actively misleading here — a model that ignores its inputs scores higher than the trained one.
- Class weighting moved the model along the precision-recall curve without lifting it; an explicit threshold chosen on validation was used instead.
- Undocumented placeholder codes carried real signal and were handled with missing-value indicators rather than imputed away.
- The gap between in-sample and held-out performance was measured and reported, not omitted.

The engineering discipline is what makes the figures trustworthy: preprocessing held inside the saved pipeline so training and inference cannot diverge, the threshold stored with the model, and every documented number generated from the artifact rather than transcribed.

Table 7.1: Technology stack

Layer	Technology
Dataset	Kaggle Give Me Some Credit — 150,000 records
Model	scikit-learn RandomForestClassifier, joblib
Data	pandas, NumPy; matplotlib, seaborn
API	FastAPI, Pydantic, Uvicorn
Frontend	React, TypeScript, Vite, Tailwind CSS
Testing	pytest — 61 tests
Deployment	Vercel (frontend and backend)

The honest conclusion is that this system extracts real, measurable signal from borrower data and is simultaneously nowhere near fit to decide anybody's credit application. Holding both statements at once is the central skill the exercise teaches.

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